

Regression:

Motivation:

So far, our study of inference has focused only on parameters (i.e. quantities) that are unknown in a population.

With one exception (Z-Sample Hypothesis Testing), we have looked at one quantity in one population.

Ex:

- Height of a person

- Weight of a person

- Likelihood of "H" in a Mystery Coin

- Avg. Blood Pressure in Treated vs Untreated Patients

↳ Always a single quantity ↳ two different populations

Sometimes, we are given several quantities for the same population (see ☆)

In such cases, a very natural question to ask is:

"What is the relationship between these quantities?"

→ e.g. What is the relationship b/w a person's height and their weight?

Note: Unlike a two-sample H. Test, we are observing both height and weight for the same population.

Z-sample H.T. $\Rightarrow$ 1 quantity, 2 populations

Current Example $\Rightarrow$ 2 quantities, 1 population

So, we're working with paired-up r.v.'s, e.g. (X_i, Y_i) for each person i

Height $\uparrow$ Weight $\uparrow$

→ We will assume that pairs (X_i, Y_i) come from some joint distribution. This joint dist'n can tell us about the relationship between X and Y .

General Setup:

From some "population", we obtain a sample $(X_1, Y_1) \dots (X_n, Y_n)$ of size n . (Often written as $(X_i, Y_i)_{i=1}^n$)

Examples:

i	X	Y
• A different precept	• Size of a Precept	• Avg. Score (within a precept) on the final
★ • A person	• Height (in.)	• Weight (lbs.)
• A country	• Cost of education	• Standard of living

What can we say about the relationship between these two quantities?

In this setup, we believe there is some relationship b/w X and Y - and we'd like to say something quantitative about it.

→ So X_i and Y_i (for any i) may not be independent.

We'd like to quantify their dependence!

We briefly talked about quantifying dependence between two r.v.'s in two ways:

1) Covariance: $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$ $\leftarrow$ Anywhere in $(-\infty, \infty)$

$\hookrightarrow$ Depends on the units of X, Y

2) Correlation: $\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$ $\leftarrow$ Must be in $[-1, +1]$

$\hookrightarrow$ Doesn't depend on units!

These are both ways of quantifying a relationship b/w X, Y .

Recall: Covariance is a linear measure, $\text{Cov}(a \cdot X, Y) = a \cdot \text{Cov}(X, Y)$ for any constant a .

This is a good start, but importantly these don't tell us:

i) anything about the conditional dist'n of Y , given $X=x$

ii) how much I expect Y to change if X changes a bit

In our example, we may want to know:

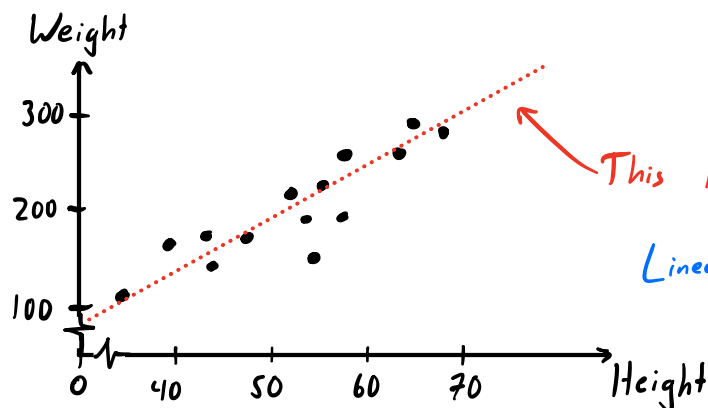
i) What is the dist'n of weight for a person with height of x ?

ii) How much does avg. weight change if height increases by Δx ?

$\hookrightarrow$ To answer these, we'll need a new tool- Linear Regression!

Main Idea:

Imagine we get a sample of people, and plot Weight vs. Height:



This line seems to "fit" the data well!

Linear Regression = Computing + Assessing the "best fit line"

Questions we'd like to answer:

- 1) What must we assume about our data?
- 2) How do we define "best fit"? How do we compute the "best fit" line?
- 3) How do we know that this "best fit" is:
 - a) Good?
 - b) Statistically Significant? (Think of Hypothesis Testing)

Regression Assumptions:

Data $(X_i, Y_i)_{i=1}^n$ are iid according to a joint distribution where:

$$Y_i = \underbrace{f(X_i)} + \underbrace{\varepsilon_i}$$

Some "noise" so that Y_i is not perfectly described by $f(X_i)$
Some function of X_i — not a density function

Note: X is called the independent variable, Y is called the dependent variable.

Here, we say we are regressing Y on X .

Aside: This is the general setup of a supervised learning problem — given $(X_i, Y_i)_{i=1}^n$, we want to learn the function $f(\cdot)$.

We will make two simplifying assumptions:

intercept slope

1) $f(X_i) = \beta_0 + \beta_1 X_i$, for some unknown pop. parameters $\beta_0, \beta_1 \in \mathbb{R}$ (Think of " θ ")

2) $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$, for some $\sigma_\varepsilon^2 > 0$, and is independent of X_i

→ Constitute a classical Linear Regression setting

Summary:

- Data given by $(X_i, Y_i)_{i=1}^n$, such that $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
- $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$ are independent of X_i 's, with unknown σ_ε^2
- β_0, β_1 are unknown — we want to estimate them using our data

An Observation: $\mathbb{E}[Y_i | X_i = x_i] = \mathbb{E}[\beta_0 + \beta_1 X_i + \varepsilon_i | X_i = x_i]$

$$= \beta_0 + \beta_1 x_i + \mathbb{E}[\varepsilon_i | X_i = x_i]$$

$$= \mathbb{E}[\varepsilon_i] \text{ by } \varepsilon_i \text{ indep. of } X_i \\ = 0$$

$$\Rightarrow \mathbb{E}[Y_i | X_i = x_i] = \underline{\beta_0 + \beta_1 x_i} \quad \left[\begin{array}{l} \text{Conditional Exp. of } Y_i \text{ is linear} \\ \text{in } x_i! \end{array} \right]$$

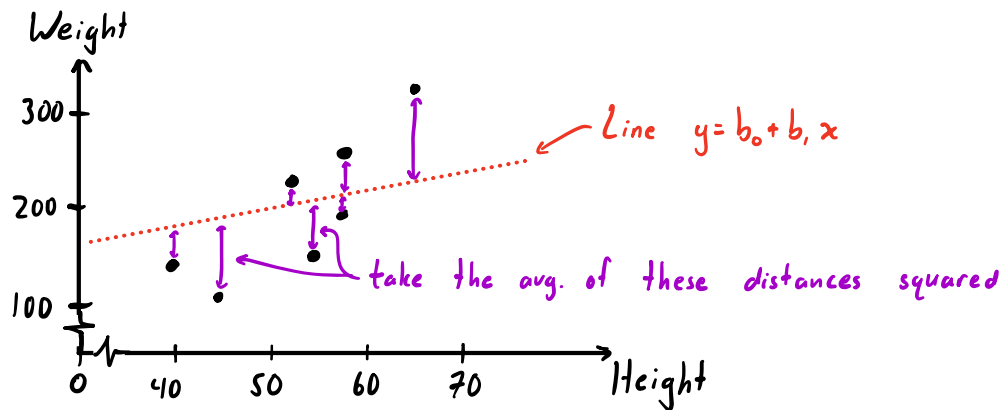
Regression Fit:

We care about minimizing the mean squared error (MSE)

Def. (MSE): For $(x_i, y_i)_{i=1}^n$, and any b_0, b_1 , the mean squared error is defined as:

$$MSE(b_0, b_1) := \underbrace{\frac{1}{n} \sum_{i=1}^n}_{\text{"mean"}} \underbrace{[y_i - (b_0 + b_1 x_i)]^2}_{\text{"error"} \text{ "squared"}}$$

Visually, using our motivating example:



Note: This is a choice - there are other measures of "best fit"

So why choose MSE?

- It is analytically "nice"
- Minimizing MSE $\Leftrightarrow$ Finding max. likelihood estimates for β_0, β_1 (Exercise)

The "best fit line" will be defined as: $y = \hat{\beta}_0 + \hat{\beta}_1 x$,

where $(\hat{\beta}_0, \hat{\beta}_1) := \underset{b_0, b_1}{\operatorname{argmin}} \underbrace{\frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2}_{= MSE(b_0, b_1)}$ (i.e. find the line, determined by intercept b_0 and slope b_1 , that minimizes M.S.E.)

Q: How do we find the best fit line?

We can use the same idea from max. likelihood, only now we have an argmax over two variables:

- $\rightarrow$ Take partial derivatives of $MSE(b_0, b_1)$ w/respect to both b_0 and b_1 .
- $\rightarrow$ Set these both equal to 0 (which occurs at $\hat{\beta}_0, \hat{\beta}_1$) and solve the system of eq's.

$$\frac{\partial MSE}{\partial \beta_0}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{2}{n} \sum_{i=1}^n [y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i] = 0 \quad (1)$$

$$\frac{\partial MSE}{\partial \beta_1}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{2}{n} \sum_{i=1}^n (-x_i)[y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i] = 0 \quad (2)$$

① re-arranges to: $\hat{\beta}_0 = \underbrace{\left(\frac{1}{n} \sum_{i=1}^n y_i\right)}_{=\bar{y}} - \hat{\beta}_1 \cdot \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i\right)}_{=\bar{x}} = \bar{y} - \hat{\beta}_1 \bar{x}$

② re-arranges to: $\hat{\beta}_0 \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i\right)}_{=\bar{x}} = \left(\frac{1}{n} \sum_{i=1}^n x_i y_i\right) - \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n x_i^2\right)$

Substitute ① into ② to isolate $\hat{\beta}_1$:

$$\bar{x} \bar{y} - \hat{\beta}_1 \bar{x}^2 = \left(\frac{1}{n} \sum_{i=1}^n x_i y_i\right) - \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n x_i^2\right)$$

$$\Downarrow$$

$$\hat{\beta}_1 \left[\underbrace{\frac{1}{n} \sum_{i=1}^n x_i^2}_{\text{variance of } x_1, \dots, x_n} - \bar{x}^2 \right] = \underbrace{\frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}}_{\text{sample covariance between } x_1, \dots, x_n \text{ and } y_1, \dots, y_n}$$

Looks like the sample
variance of $x_1, \dots, x_n$

Looks like the sample covariance
between $x_1, \dots, x_n$ and $y_1, \dots, y_n$

Some useful notation:

$$\frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{S_{xx}}{n}$$

$$\frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{S_{xy}}{n}$$

Both of these follow by re-arranging

$$\therefore \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Note: Minimizing the mean squared error is "nice" because we get closed-form solutions for $\hat{\beta}_0$ and $\hat{\beta}_1$. If we were to use a different notion of "fit", we may not!

Quality of Fit:

In practice, our "best fit line" will never perfectly fit the data.

So, we'd like to assess how "good" the fit is.

Def. (Fitted Values, Residuals): For $\hat{\beta}_0$ and $\hat{\beta}_1$, denote:

- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ as the **fitted value** for each i
- $\hat{\epsilon}_i = y_i - \hat{y}_i$ as the **residual** of i

Note: The residuals $\hat{\epsilon}_i$ are not necessarily the errors ϵ_i .

The errors are $\epsilon_i = y_i - \beta_0 - \beta_1 x_i$, which we don't know since both β_0 and β_1 are unknown.

Def. (Sum of Squared Errors, SSE): The **sum of squared errors**, also called the "**residual sum of squares**" is:

$$SSE = n \cdot MSE(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^n \hat{\epsilon}_i^2$$

Note: The SSE quantifies how much variation in $y_1, \dots, y_n$ is not explained by the best fit line $\hat{\beta}_0, \hat{\beta}_1$.

Note: Recall that the variance of the errors ϵ_i is σ_ϵ^2 , and is unknown.

It turns out that a good estimate of σ_ϵ^2 is

$$\hat{\sigma}_\epsilon^2 \text{ is called the } \underline{\text{residual standard error}} \quad \hat{\sigma}_\epsilon^2 = \frac{SSE}{n-2}$$

↑ This "-2" comes from already having estimated two unknown quantities to compute this: $\hat{\beta}_0$ and $\hat{\beta}_1$.

This estimate is "good" in that the estimator (before observing data) is unbiased for σ_ϵ^2 .

SSE is a good start, but it isn't "normalized" - if, for instance, we add an extra data point to the sample, then SSE may increase even if the "fit" is better.

Def. (Total Sum of Squares, SST): The total sum of squares is:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2 = S_{yy}$$

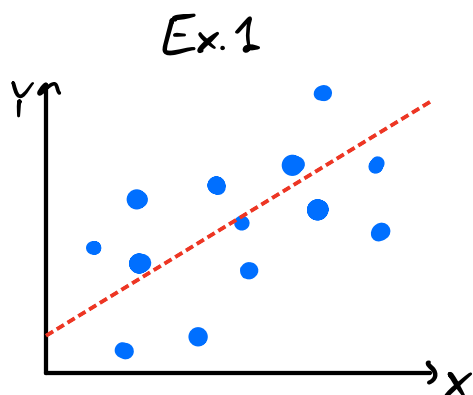
Note: SST is telling us how much variation there is in $y_1, \dots, y_n$.
So combining this with SSE will tell us what fraction of the variation in $y_1, \dots, y_n$ is not explained by the best fit line!

Def. (Coefficient of Determination, r^2): The coefficient of determination is:

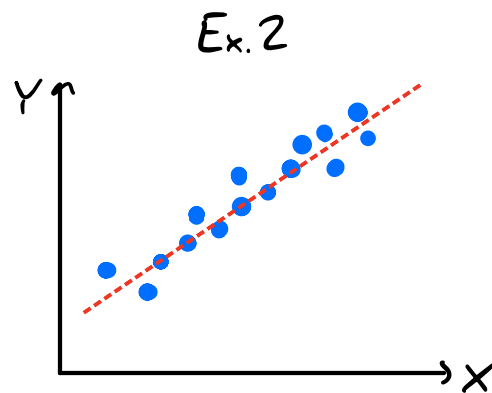
$$r^2 := 1 - \frac{SSE}{SST} \in [0, 1]$$

Note: If our model is "good", then SSE is small relative to SST , and r^2 is large (close to 1). If our model is bad, it doesn't explain much of the variation in $y_1, \dots, y_n$, and r^2 is close to 0.

Visually:



- Low r^2
- Intuitively doesn't look like a good fit...



- High r^2
- Seems like a great fit!

Properties of Estimators:

Before we get our data $(x_i, y_i)_{i=1}^n$, our estimates $\hat{\beta}_0, \hat{\beta}_1$ would be random.

Equivalently, if we obtained many different samples, we would have estimates $\hat{\beta}_0, \hat{\beta}_1$ for each.

→ We want to understand the properties of $\hat{\beta}_0$ and $\hat{\beta}_1$ as (random) point estimators

Recall: We assumed that

- $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
- $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$ is independent of X_i

Prop. (Properties of Linear Reg. Estimators): Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the (random) best fit line for (random) data $(X_i, Y_i)_{i=1}^n$. If all of the assumptions are true, then:

- Unbiased: $E[\hat{\beta}_0] = \beta_0$, $E[\hat{\beta}_1] = \beta_1$
- Variance: Given $x_1, \dots, x_n$, $Var(\hat{\beta}_1 | X_1, \dots, X_n = x_1, \dots, x_n) = \frac{\sigma_\varepsilon^2}{S_{xx}} = \left(\underbrace{SE(\hat{\beta}_1)}_{\text{"Standard Error of } \hat{\beta}_1}} \right)^2$
- Sampling Distribution: $\frac{\hat{\beta}_1 - \beta_1}{\frac{\hat{\sigma}_\varepsilon}{\sqrt{S_{xx}}}} \sim t_{n-2}$
 $\hat{\sigma}_\varepsilon$ "Residual Std. Error" = $\sqrt{\frac{SSE}{n-2}}$

Note: These will allow us to do inference about the unknown β_1

→ What is a 95% LI for β_1 ?

→ Is β_1 statistically different from 0 at 5% level?

Proof: Unbiased

i) $\hat{\beta}_1$:

$$E[\hat{\beta}_1] = E\left[\frac{S_{xy}}{S_{xx}}\right]$$

$$= E\left[\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

by def'n

$$= E\left[\frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_0 + \beta_1 X_i + \varepsilon_i - \beta_0 - \beta_1 \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, and
 $\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{\varepsilon}$, by ass'n
 $\bar{\varepsilon} = 0$

$$= E\left[\frac{\sum_{i=1}^n (X_i - \bar{X}) \cdot [\beta_1 (X_i - \bar{X}) + \varepsilon_i]}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

Simplifying

$$= E\left[\frac{\cancel{\sum_{i=1}^n (X_i - \bar{X})^2}}{\cancel{\sum_{i=1}^n (X_i - \bar{X})^2}} \cdot \beta_1\right] + E\left[\frac{\sum_{i=1}^n (X_i - \bar{X}) \varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

Linearity of $E[\cdot]$

$$= \beta_1 + \sum_{i=1}^n E\left[\frac{(X_i - \bar{X}) \varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

Since ε_i is independent of X_i 's.

$$= \beta_1 + \sum_{i=1}^n E\left[\frac{(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}\right] \cdot \underbrace{E[\varepsilon_i]}_{=0 \text{ by assumption}}$$

$$\Rightarrow \underline{E[\hat{\beta}_1] = \beta_1}$$

ii) $\hat{\beta}_0$:

$$E[\hat{\beta}_0] = E[\bar{Y} - \hat{\beta}_1 \bar{X}]$$

by def'n

$$= E[\beta_0 + \beta_1 \bar{X} - \hat{\beta}_1 \bar{X}]$$

linearity of $E[\cdot]$

$$= \beta_0 + E[(\beta_1 - \hat{\beta}_1) \bar{X}]$$

$$= E[\bar{X} \cdot \underbrace{E[\beta_1 - \hat{\beta}_1 | X_1, \dots, X_n]}_{=0, \text{ similar proof to above}}]$$

by the tower rule

$$\Rightarrow \underline{E[\hat{\beta}_0] = \beta_0} \quad \square$$

Proof: Variance of $\hat{\beta}_1$

From the earlier proof, $\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X}) \varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$

$$\text{Var}(\hat{\beta}_1 | X_1, \dots, X_n = x_1, \dots, x_n) = \text{Var}\left(\underbrace{\beta_1}_{\text{constant}} + \frac{\sum_{i=1}^n (X_i - \bar{X}) \varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid \underbrace{X_1, \dots, X_n = x_1, \dots, x_n}_{\text{plug these in}}\right)$$

$$= \text{Var}\left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sum_{i=1}^n (x_i - \bar{x}) \varepsilon_i\right)$$

pull out and square pull out since ε_i 's are indep., and $x_i - \bar{x}$ is not random

$$= \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \cdot \sum_{i=1}^n \text{Var}(\underbrace{(x_i - \bar{x}) \varepsilon_i}_{\text{pull out and square}})$$

$$= \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \cdot \sum_{i=1}^n (x_i - \bar{x})^2 \underbrace{\text{Var}(\varepsilon_i)}_{= \sigma_\varepsilon^2 \text{ for all } i}$$

$$= \sigma_\varepsilon^2 \cdot \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \cdot \sum_{i=1}^n \cancel{(x_i - \bar{x})^2}$$

$$\therefore \text{Var}(\hat{\beta}_1 | X_1, \dots, X_n = x_1, \dots, x_n) = \frac{\sigma_\varepsilon^2}{S_{xx}} \quad \square$$

Proof: Sampling Distribution

Beyond the scope of this course...

Summary:

Assumptions:

- 1) Data is iid from $Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$
↳ "There is actually a linear relationship b/w X and Y "
- 2) Errors are indep. of X_i 's, and with mean 0
↳ "Exogeneity / Exogenous Errors"
- 3) Errors $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \rightarrow$ "Normally distributed errors"

Objective: Find $(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0, b_1}{\operatorname{argmin}} \operatorname{MSE}(b_0, b_1)$

Results:

- $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- a) Estimates $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x}$.
 - b) Quality of Fit $r^2 = 1 - \frac{SSE}{SST}$
↳ $n \cdot \operatorname{MSE}(\hat{\beta}_0, \hat{\beta}_1)$
↳ $n \cdot \operatorname{MSE}(\bar{y}, 0)$
 - c) Properties of $\hat{\beta}_1$ Estimator:
 - $E[\hat{\beta}_1] = \beta_1$, $\operatorname{Var}(\hat{\beta}_1 | X_1 = x_1, \dots, X_n = x_n) = \frac{\sigma_\varepsilon^2}{S_{xx}}$
 - $\frac{\sqrt{S_{xx}}(\hat{\beta}_1 - \beta_1)}{\hat{\sigma}_\varepsilon} \sim t_{n-2}$ [We did not prove this]
↳ $\hat{\sigma}_\varepsilon = \sqrt{\frac{SSE}{n-2}}$

Next Q's to Answer:

- How do we do CI's and Hypothesis Testing for β_1 ?
 - What if I have another independent variable? I.e. $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \varepsilon_i$
 - What if the relationship b/w X and Y isn't linear? I.e. $Y_i = f(X_i) + \varepsilon_i$ for general $f(\cdot)$?
 - What can go wrong? I.e. what if our assumptions aren't true?
-